

Notes: Boutros, Clara & Gomes (JFE, 2024)
Borrow now, pay even later: A quantitative analysis of student debt payment plans

Contents

1	Big picture	3
2	Data and measurement	4
2.1	Student debt facts	4
2.2	Baseline repayment contracts	4
2.3	Alternative contracts	4
2.4	Key outcomes	4
2.5	Calibration targets and data sources	5
3	Models and empirical specifications	6
3.1	Standard Repayment Plan	6
3.2	Income-Driven Repayment Plan	6
3.3	Delinquency	6
3.4	Budget constraint and debt evolution	7
3.5	Income process and discretionary income	7
3.6	Household value function	7
3.7	Principal Payment Deferral	7
3.8	Full Payment Deferral	8
3.9	Equilibrium loan repricing	8
4	Identification and quantitative design	9
4.1	What identifies the main result	9
4.2	Why lifecycle timing matters	9
4.3	Why the lender-NPV condition matters	9
4.4	Robustness logic	9
4.5	Main limitations	10
5	Tables and figures with expanded interpretation	11
5.1	Figure 1: Student debt in the United States	11
5.2	Table 1 + Table 2: contract parameters and baseline calibration	11
5.3	Table 3 + Table 4: lifecycle repayment behavior under baseline contracts	12
5.4	Table 5 + Figure 2: alternative contracts and repayment choices	13
5.5	Table 6 + Figure 3: consumption, wealth, savings, and debt	14
5.6	Table 7 + Table 8: welfare gains and robustness	15
5.7	Table 9 + Table 10: extensions and policy comparison	17

6	Short summary	19
7	Conclusion	19
7.1	Core conclusion	19
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** Can changing the timing of student loan payments generate large welfare gains without the fiscal cost of outright debt forgiveness?
- **Motivation.** Student debt is the second largest component of U.S. consumer debt. Borrowers are required to repay early in life, when income and liquid wealth are low and marginal utility of consumption is high.
- **Main policy idea.** Replace front-loaded payments with contracts that defer payments for the first ten years of the loan.
- **Two alternative contracts.** Under **Principal Payment Deferral** (PPD), borrowers pay only interest for ten years and begin principal repayment in year 11. Under **Full Payment Deferral** (FPD), borrowers make no payments for ten years and accrued interest is added to principal.
- **Baseline model.** A quantitative lifecycle consumption-savings model with student debt, risky labor income, SRP/IDRP choices, delinquency, utility costs of IDRP enrollment and delinquency, and equilibrium repricing of loan premia.
- **Main welfare results.** Certainty-equivalent annual consumption gains are **1.22%** under PPD and **2.12%** under FPD.
- **Delinquency results.** Delinquencies fall by about one-third under PPD and one-half under FPD.
- **Mechanism.** Payment deferral raises consumption and saving early in life, when borrowers are most constrained, and it allows better insurance against income shocks.
- **Loan repricing channel.** Lower delinquency allows loan premia to fall. This channel matters for PPD, but the FPD welfare gains are driven mostly by payment deferral itself.
- **Debt forgiveness comparison.** The model-implied welfare gains are comparable to those from debt forgiveness: about \$10,800 for PPD and \$16,750 for FPD, but the deferral policies are budget-neutral in the model.
- **Economic takeaway.** Liquidity timing matters. Well-designed deferral contracts can deliver much of the benefit of student debt forgiveness without requiring a large government transfer.

2 Data and measurement

2.1 Student debt facts

- Total U.S. student debt rose from roughly \$500 billion in 2008 to almost \$1.8 trillion in 2023.
- Student debt is about 7% of GDP, up from roughly 4% in the early 2000s.
- The number of people with student debt rose strongly after 2008 and then flattened.
- Average debt per borrower rose from roughly \$18,000 in 2008 to about \$37,000 by 2023.

Interpretation: These facts motivate the policy question. Student debt is both large in aggregate and concentrated among young households that are likely to face high marginal utility of consumption.

2.2 Baseline repayment contracts

- The model starts all borrowers in a **Standard Repayment Plan** (SRP).
- Borrowers may switch into an **Income-Driven Repayment Plan** (IDRP), which caps payments at a fraction of discretionary income.
- Borrowers may also skip payments and become temporarily delinquent, paying a utility cost and accumulating missed interest.
- The SRP has a 10-year amortization period.
- The IDRP has a maximum maturity of 25 years, after which remaining principal is discharged without penalty.

Interpretation: The baseline economy includes the two main repayment choices seen in the data, plus delinquency. This matters because deferral contracts are compared against an economy where borrowers already have some payment flexibility through IDRP.

2.3 Alternative contracts

- **PPD: Principal Payment Deferral.** Borrowers pay only interest during the first ten years. Principal payments begin later.
- **FPD: Full Payment Deferral.** Borrowers make no payments during the first ten years. Interest accrues and is added to principal.
- The policies are evaluated in counterfactual equilibria: households re-optimize and loan premia are repriced so the lender's expected net present value is unchanged.

Interpretation: PPD is a partial liquidity policy; FPD is a stronger liquidity policy. Both move required payments from young ages to later ages.

2.4 Key outcomes

- Repayment-plan shares: SRP, IDRP, delinquency.
- Debt outstanding and debt-to-income ratios.
- Consumption, consumption growth, savings, and net wealth over the lifecycle.
- Delinquency rates and negative amortization events.

- Lender cash flows and equilibrium loan premia.
- Welfare, measured as annual certainty-equivalent consumption gains.

Interpretation: The paper is not only asking whether borrowers pay less early in life. It asks whether timing changes improve lifecycle consumption smoothing and reduce delinquency while maintaining lender break-even.

2.5 Calibration targets and data sources

- Income and wealth moments are calibrated using Survey of Consumer Finances data.
- Income dynamics use estimates from Guvenen et al. (2021) and Cocco et al. (2005).
- The model targets observed shares of households on SRP, on IDRP, or delinquent.
- The utility cost of IDRP enrollment and the utility penalty for delinquency are calibrated to match these shares.

Interpretation: The model is disciplined by both repayment-plan behavior and household balance-sheet moments, not just by average debt levels.

3 Models and empirical specifications

3.1 Standard Repayment Plan

The standard plan amortizes the initial loan balance S_0 over N_{SRP} periods. The total payment is

$$P_t^{\text{SRP}} + I_t^{\text{SRP}} = \left[\frac{1 - (1 + r_s)^{-N_{\text{SRP}}}}{r_s} \right]^{-1} S_0, \quad (1)$$

where $r_s = r_f + \varphi^{\text{Baseline}}$ is the student-loan interest rate.

Interest and principal are

$$I_t^{\text{SRP}} = r_s S_t, \quad (2)$$

$$P_t^{\text{SRP}} = \left[\frac{1 - (1 + r_s)^{-N_{\text{SRP}}}}{r_s} \right]^{-1} S_0 - r_s S_t. \quad (3)$$

Interpretation: In SRP, payments are constant but early payments mostly cover interest. The debt burden is therefore front-loaded in consumption terms because borrowers pay when income and wealth are low.

3.2 Income-Driven Repayment Plan

The IDR plan caps total payments at a fraction of discretionary income:

$$P_t^{\text{IDRP}} + I_t^{\text{IDRP}} = \min\{P_t^{\text{SRP}} + I_t^{\text{SRP}}, \omega_{\text{IDRP}} \text{DI}_t\}. \quad (4)$$

Interest is still calculated on the outstanding student debt:

$$I_t^{\text{IDRP}} = r_s S_t, \quad (5)$$

$$P_t^{\text{IDRP}} = \min\{P_t^{\text{SRP}} + I_t^{\text{SRP}}, \omega_{\text{IDRP}} \text{DI}_t\} - I_t^{\text{IDRP}}. \quad (6)$$

If the income-based payment does not cover interest,

$$\omega_{\text{IDRP}} \text{DI}_t < r_s S_t, \quad (7)$$

then principal repayment is negative and the outstanding balance grows.

Interpretation: IDR provides liquidity insurance through lower payments, but it can create negative amortization and lengthen repayment.

3.3 Delinquency

If the household becomes delinquent, it skips the payment in exchange for a utility penalty. The missed interest is added to the balance:

$$P_t^D = -I_t^D, \quad (8)$$

$$S_{t+1} = (1 + r_s) S_t. \quad (9)$$

Interpretation: Delinquency gives temporary liquidity but worsens future debt. It is a costly state that captures payment distress without discharge of the student loan.

3.4 Budget constraint and debt evolution

Household wealth evolves according to

$$W_{t+1} = (1 + r_f)(W_t - C_t^j - P_t^j - I_t^j) + (1 - h_{t+1} - \tau)Y_{t+1}, \quad (10)$$

where $j \in \{\text{SRP}, \text{IDRP}, D\}$.

If the household is not delinquent, student debt evolves as

$$S_{t+1} = S_t - P_t^j. \quad (11)$$

Interpretation: The budget constraint captures the central friction: student-loan payments reduce liquid resources early in life, exactly when households also have low wealth and income risk.

3.5 Income process and discretionary income

Working-life income is

$$Y_t^i = (1 - \nu_t^i) \exp(g(t) + \alpha_i + z_{it} + \epsilon_{it}), \quad (12)$$

with persistent income

$$z_{it} = \rho z_{it-1} + \eta_{it}. \quad (13)$$

Discretionary income for IDRP is

$$\text{DI}_t = \exp(g(t) + \alpha + z_t) - \omega_{FPL} \times FPL. \quad (14)$$

Interpretation: The income process creates lifecycle income growth and idiosyncratic shocks. This is why deferring student-loan payments can improve smoothing.

3.6 Household value function

Each period, the household chooses among SRP, IDRP, and delinquency:

$$V_t(W_t, S_t, z_t) = \max \{ V_t^{\text{SRP}}(W_t, S_t, z_t), V_t^{\text{IDRP}}(W_t, S_t, z_t), V_t^D(W_t, S_t, z_t) \}. \quad (15)$$

The auxiliary value functions have Epstein-Zin preferences and include utility costs ξ^j for IDRP or delinquency choices.

Interpretation: The discrete choice captures payment-plan switching, while the continuous choice captures consumption-savings behavior.

3.7 Principal Payment Deferral

Under PPD, for the first N_{PPD} periods the household pays only interest:

$$I_t^{\text{PPD}} = r_s^{\text{PPD}} S_t, \quad (16)$$

$$P_t^{\text{PPD}} = 0. \quad (17)$$

After the deferral period, the contract reverts to the standard structure using the remaining balance.

Interpretation: PPD lowers required payments early in life while preventing the balance from growing mechanically when the household is not delinquent.

3.8 Full Payment Deferral

Under FPD, for the first N_{FPD} periods borrowers make no payments:

$$I_t^{\text{FPD}} = r_s^{\text{FPD}} S_t, \quad (18)$$

$$P_t^{\text{FPD}} = -I_t^{\text{FPD}}. \quad (19)$$

The balance after the initial deferral period is

$$S_{N_{\text{FPD}}}^{\text{FPD}} = (1 + r_s^{\text{FPD}})^{N_{\text{FPD}}} S_0. \quad (20)$$

Interpretation: FPD maximizes early-life liquidity but increases debt outstanding, shifting repayment later in life.

3.9 Equilibrium loan repricing

For each counterfactual contract, the model solves a new equilibrium loan premium so that the lender's expected net present value is unchanged relative to the baseline.

Interpretation: This is crucial. The welfare gains are not driven by a subsidy from lenders or taxpayers; they are gains from changing payment timing and reducing delinquency.

4 Identification and quantitative design

4.1 What identifies the main result

- The model is calibrated to the current student-loan environment and then used for counterfactual policy experiments.
- The central comparison is between the baseline equilibrium and new equilibria under PPD and FPD.
- Households re-optimize in each policy environment.
- Loan premia are repriced to hold lender expected NPV fixed.

Interpretation: The exercise is a structural policy counterfactual, not a reduced-form causal design. Its credibility depends on matching key baseline moments and on the robustness of welfare gains across alternative assumptions.

4.2 Why lifecycle timing matters

- Young households have low liquid wealth and high marginal utility of consumption.
- Current student-loan contracts force repayments early in life.
- Deferral moves required payments to later periods when households generally have higher income and more wealth.
- This improves both intertemporal smoothing and insurance against income shocks.

Interpretation: The mechanism is liquidity timing, not debt cancellation.

4.3 Why the lender-NPV condition matters

- PPD and FPD change the timing and risk of cash flows to lenders.
- Lower delinquency reduces expected nonpayment and can lower the equilibrium loan premium.
- The authors solve for new loan premia so the expected NPV of the loan is held constant.

Interpretation: This makes the policies budget-neutral in the model and allows them to be compared with debt forgiveness.

4.4 Robustness logic

The paper checks robustness to:

- a higher NPV requirement for lenders;
- different preference parameters;
- heterogeneous initial student-debt balances;
- access to unsecured credit;
- different household compositions;
- endogenous stock-market participation;
- prepayment options;
- Biden administration proposals on debt forgiveness and IDR reform.

Interpretation: The welfare gains remain economically meaningful under these alternatives, suggesting that the liquidity-timing mechanism is robust.

4.5 Main limitations

- The baseline model abstracts from housing purchase decisions, job search, family planning, portfolio choice, and 401(k) employer matching.
- The model abstracts from endogenous education choices and the supply response of colleges.
- Credit score effects are not explicitly valued.
- Labor supply is exogenous in the baseline, although an extension with endogenous labor supply gives larger welfare gains.
- The model focuses on representative repayment-policy design rather than administrative or political implementation.

Interpretation: These omissions make the measured welfare gains conservative in several dimensions.

5 Tables and figures with expanded interpretation

5.1 Figure 1: Student debt in the United States

Read:

- Panel A shows total student debt rising from roughly \$500 billion in 2006 to almost \$1.8 trillion by 2021.
- Panel B shows student debt rising relative to GDP, so the growth is not simply an artifact of the overall economy becoming larger.
- Panel C shows that the number of borrowers rises and then flattens, which implies that recent growth is not only from more borrowers entering the system.
- Panel D shows average debt per borrower rising from below \$20,000 to above \$35,000, which means the intensive margin matters.

Interpretation:

- The four panels together show that student debt growth has both an aggregate and household-level dimension. Policy design must therefore consider not only the number of borrowers, but also the repayment burden faced by each borrower.
- The flattening of borrower counts combined with rising average debt per borrower motivates a contract-design approach. If borrowers owe more per person, the timing of repayment becomes more important even when the number of borrowers is stable.
- The figure also clarifies why the paper focuses on liquidity and lifecycle timing rather than only total debt cancellation: the central stress is that a large stock of debt is concentrated early in life.

Economic message: Student debt is a large and growing household balance-sheet item. The figure turns the paper’s policy question from an abstract contract-design problem into a quantitatively relevant U.S. household-finance problem.

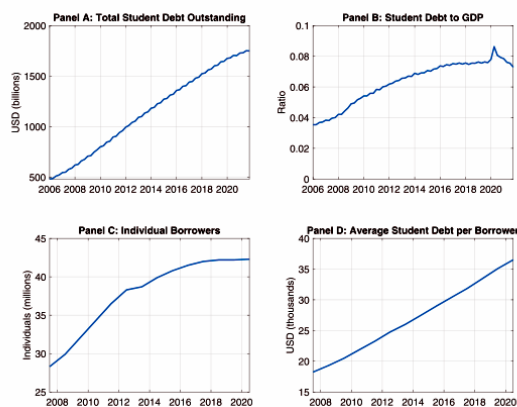


Figure 1: Student debt in the United States. Directly cropped from the paper.

5.2 Table 1 + Table 2: contract parameters and baseline calibration

Read:

- Table 1 sets the baseline student-loan premium at 3.5%, the SRP length at 10 years, and the IDR maximum length at 25 years.

- IDRPs payments use 10% of discretionary income and discretionary income is calculated relative to 150% of the federal poverty level.
- Table 2 calibrates the utility costs of IDRPs enrollment and delinquency.
- The model matches targeted repayment-state shares closely: the data have 50.7% SRP, 31.3% IDRPs, and 18.0% delinquency; the model gives 56.1%, 27.9%, and 16.0%.
- Untargeted moments are qualitatively consistent: delinquent households have lower income, higher debt-to-income ratios, and lower wealth.

Interpretation:

- Table 1 tells us what the baseline contract actually is. This is important because the welfare gains from PPD and FPD are measured relative to a baseline with substantial existing flexibility through IDRPs, not relative to a world with only rigid standard repayment.
- The IDRPs utility cost in Table 2 is not a literal application fee. It is a reduced-form way to match the fact that many eligible borrowers do not use IDRPs. It captures hassle costs, lack of information, psychological costs, paperwork, and other frictions.
- The delinquency penalty is also reduced form. It is chosen so the model reproduces observed delinquency behavior, and then counterfactual policies ask how delinquency changes when required payments are moved later.
- The untargeted rows are crucial for credibility. If the model matched repayment shares but made delinquent households too rich or too low-debt, it would not be credible for welfare analysis. The table shows the model broadly preserves the ordering of repayment states by income, debt burden, and wealth.

What not to over-read:

- The table does not prove that the calibrated utility costs are structural constants. They are parsimonious wedges that discipline repayment choices in the model.
- The calibration is designed for counterfactual contract evaluation, not for estimating the exact administrative cost of IDRPs or the exact psychological cost of delinquency.

Economic message: The baseline model replicates repayment-state behavior well enough to evaluate counterfactual contracts. The model is credible because it matches both choices and balance-sheet patterns.

Table 1
Repayment plan parameters: This table shows parameters related to the repayment plans in the model.

Parameter		Value
Interest rate premium on student debt	φ	0.035
Standard repayment plan length	N_{SRP}	10
Income-driven repayment plan maximum length	N_{IDRP}	25
Fraction of Federal Poverty Level (FPL) for constructing discretionary income	ω_{DI}	1.50
Fraction of discretionary income for IDRPs payment	ω_{IDRP}	0.10

Table 1: repayment plan parameters.

Table 2
Model and data: Panel A shows the calibrated utility costs of delinquency and IDRPs enrollment, which are chosen to match the model and data moments in Panel B. Panel C compares four untargeted moments: average loan balances, average income, debt to income, and net wealth. Debt to income is student debt to income, while net wealth is the overall net wealth of the household. All data moments from the SCF.

Panel A: Calibrated parameters

Parameter	Value	Source/Target
IDRP utility penalty	ζ^{IDRP}	0.00001
Delinquency utility penalty	ζ^D	0.0345

Panel B: Targeted moments

Moment	Data	Model
Proportion of SRP	0.507	0.561
Proportion of IDRPs	0.313	0.279
Proportion of delinquency	0.180	0.160

Panel C: Untargeted moments

Moment	Data			Model		
	SRP	IDRP	Delinquency	SRP	IDRP	Delinquency
Median loan balance	14,000	22,000	24,000	14,291	26,109	32,598
Median income	76,359	55,996	30,544	80,416	45,000	18,303
Median debt-to-income ratio	0.179	0.426	0.655	0.170	0.574	1.769
Median net wealth	10,480	-3,790	-10,470	14,650	-12,551	-28,197

Table 2: calibration and model fit.

5.3 Table 3 + Table 4: lifecycle repayment behavior under baseline contracts

Read:

- Table 3 breaks down SRP, IDRPs, and delinquency by age group.

- Younger borrowers rely heavily on IDR; delinquency becomes especially common among older households that still have debt.
- IDR lowers payments relative to SRP, but can leave larger balances outstanding.
- Table 4 shows that negative amortization is rare for IDR but delinquency produces no payment and increases the debt balance.
- Delinquency is persistent: borrowers who are delinquent are much more likely to be delinquent again next period.

Interpretation:

- Table 3 is the baseline diagnostic table. It shows what problem the new contracts are trying to solve: repayment burdens are concentrated among people who are young, income-constrained, or still carrying debt unusually late in life.
- The age decomposition matters because average statistics can hide lifecycle stress. A repayment contract that looks affordable on average may be inefficient if payments arrive before income and wealth have accumulated.
- The IDR columns show that existing income-driven policy provides insurance, but not enough. Borrowers still end up in delinquency, and some remain in repayment states later in life.
- Table 4 deepens the interpretation by showing how debt evolves conditional on payment state. Delinquency is not only a missed-payment event; it is a state in which debt can grow, creating persistence and making future repayment harder.
- Together, Tables 3 and 4 show that delinquency is both a symptom of liquidity stress and a mechanism that worsens future balance sheets.

What not to over-read:

- The tables should not be read as evidence that IDR is ineffective. IDR reduces payments for constrained households. The point is that IDR alone does not fully solve the timing mismatch between repayment obligations and early-life resources.

Economic message: The baseline contracts provide some insurance, but they do not eliminate early-life repayment stress or later-life high balances among distressed borrowers. This is the core reason PPD and FPD can generate welfare gains.

Table 3
Standard repayment plan vs. income-driven repayment plan during the lifecycle: This table shows average income, income growth, consumption, consumption growth, probability of being enrolled in SRP/IDR/Delinquency, average debt outstanding, average debt payments, average net wealth, and average debt payments to income for households during their working age over the lifecycle. Panel A reports the statistics when agents choose the SRP, Panel B reports the same when agents choose the IDR, and the last panel when agents choose to become delinquent. All statistics are conditional on the agent having debt outstanding (with the exception of the second line of the table). On the second line we report the probability that agents choose SRP, choose IDR, or become delinquent, conditional on having debt outstanding at age 25.

	Panel A: Enroll in SRP				Panel B: Enroll in IDR				Panel C: Delinquency			
	26-30	31-35	36-40	41-45	26-30	31-35	36-40	41-45	26-30	31-35	36-40	41-45
Probability	0.435	0.674	0.691	0.304	0.501	0.212	0.105	0.204	0.064	0.114	0.204	0.492
Probability (as a fraction of age 25)	0.435	0.674	0.352	0.018	0.501	0.212	0.053	0.012	0.064	0.114	0.104	0.030
Income	73.272	96.354	92.056	81.301	42.147	45.254	42.893	43.943	16.512	20.886	21.056	21.910
Income growth	0.130	0.078	0.048	0.058	0.059	0.011	0.005	0.000	-0.119	-0.018	0.016	0.002
Consumption	43.080	55.619	54.520	47.272	23.450	25.779	25.059	26.616	12.740	14.721	13.788	13.127
Consumption growth	0.137	0.084	0.058	0.071	0.052	0.014	0.005	-0.003	-0.070	-0.001	0.014	0.002
Leverage outstanding	24.654	13.670	6.298	19.458	26.947	20.474	17.612	33.960	28.823	29.385	32.829	43.134
Net wealth	-7.253	22.696	41.755	39.598	-14.708	-1.588	8.780	1.685	-27.157	-24.721	-25.458	-34.482
Total payments	3.610	3.610	3.031	3.271	2.803	3.154	3.185	2.936	0.000	0.000	0.000	0.000
Principal payments	2.503	2.995	2.748	2.396	1.591	2.232	2.392	1.408	-1.297	-1.322	-1.477	-1.941
Interest payments	1.107	0.615	0.283	0.875	1.213	0.921	0.793	1.528	1.297	1.322	1.477	1.941
Debt payments to debt outstanding	0.148	0.299	0.778	0.405	0.106	0.166	0.235	0.168	0.000	0.000	0.000	0.000
Debt payments to income	0.052	0.044	0.042	0.046	0.070	0.077	0.085	0.078	0.000	0.000	0.000	0.000

Table 4
Determinants of debt repayments: This table shows the determinants of debt repayments. The two left-hand columns show agents' choices that lead to positive amortization of debt (either payments under SRP or IDR) whereas the two right-hand columns show agents' choices that lead to negative amortization (either IDR or delinquency). The first row shows the probability. The second and third rows show debt to income and debt payments to income. The fourth, fifth and sixth rows show age, income, and net wealth respectively. The last four rows show principal and interest payments (realized and counterfactual payments under the SRP).

	Positive amortization ($\Delta S_{t+1} > 0$)		Negative amortization ($\Delta S_{t+1} < 0$)	
	SRP	IDR	IDR	Delinquency
Probability	0.561	0.269	0.010	0.160
Debt-to-income	0.201	0.668	2.006	1.970
Debt payments to income	0.045	0.075	0.055	0.000
Principal payment	2.781	1.902	-0.446	-1.577
Interest payment	0.668	1.116	1.350	1.577
Comparable principal payment under SRP	2.779	3.795	2.260	2.033
Comparable interest payment under SRP	0.668	1.116	1.350	1.577

5.4 Table 5 + Figure 2: alternative contracts and repayment choices

Read:

- Table 5 compares the baseline economy with PPD and FPD.

- Under the alternative contracts, more households use SRP-like payment behavior and fewer become delinquent.
- Under PPD, the overall delinquency share falls from 16.0% to 11.1%.
- Under FPD, delinquency falls further to 9.7%.
- Figure 2 visually shows the repayment-state shares: SRP rises from 56% in the baseline to about 80% under PPD and 81% under FPD, while delinquency falls sharply.

Interpretation:

- The main lesson is not that borrowers suddenly become unconstrained forever. The lesson is that removing required early payments makes borrowers less likely to use costly fallback options.
- The rise in SRP-like behavior under PPD and FPD is economically meaningful. It says that once the contract is better timed, many households do not need to rely on IDR or delinquency as much.
- The difference between PPD and FPD is informative. PPD still requires interest payments, so it relaxes liquidity constraints only partly. FPD removes payments completely during the deferral window, which explains why delinquency falls more under FPD.
- Figure 2 is a compact visualization of a central mechanism: payment deferral changes the composition of repayment states, not just the level of payments.

What not to over-read:

- FPD's lower delinquency rate should not be interpreted as "free." It comes with higher debt balances later because interest accrues and is capitalized. The welfare question is whether this later cost is worth the early-life liquidity benefit.

Economic message: Deferring payments reduces the need to use delinquency as a liquidity-management tool. FPD is stronger than PPD because it attacks the payment-timing problem more directly.

Table 5
Debt repayment, wealth, income, and consumption in the three different economies: This table shows average probability of agents being enrolled in the SRP/IDRP/Delinquency, conditional on having debt outstanding (first three rows), on taking a loan at age 25 (fourth to sixth rows), and on income, consumption, average debt outstanding, average net wealth, average debt payments, average debt payments to debt outstanding, and average debt payments to income for households during their working age over the lifecycle. Panel A shows the statistics for the baseline case, and Panels B and C show the statistics for the two student debt restructuring policies under analysis: PPD and FPD, respectively. For the first 10 years of the lifecycle, agents only make interest payments under the PPD policy (Panel B), or defer payments for 10 years under the FPD policy (Panel C).

	Panel A: Baseline				Panel B: PPD				Panel C: FPD			
	Overall	26-35	36-40	41-65	Overall	26-35	36-40	41-65	Overall	26-35	36-40	41-65
Pct of agents on SRP	0.561	0.568	0.691	0.304	0.800	0.970	0.752	0.619	0.806	1.000	0.752	0.661
Pct of agents on IDR	0.279	0.341	0.105	0.204	0.089	0.000	0.132	0.170	0.096	0.000	0.126	0.168
Pct of agents delinquent	0.160	0.092	0.204	0.492	0.111	0.030	0.115	0.210	0.097	0.000	0.122	0.172
Income	65.360	66.759	72.799	44.444	80.863	66.759	96.431	87.716	85.320	66.759	96.431	96.346
Consumption	38.323	38.705	43.523	27.265	49.158	39.567	57.986	55.046	52.619	39.971	57.962	61.234
Leverage outstanding	20.909	20.968	12.904	34.071	23.846	28.640	24.430	17.416	26.772	31.559	31.568	20.140
Net wealth	5.132	1.293	24.467	-4.603	30.395	-1.109	44.676	60.004	38.206	-3.066	40.369	73.869
Payments	2.757	3.045	2.427	1.593	1.824	0.743	2.889	2.439	1.656	0.000	2.840	2.545
Principal payments	1.816	2.102	1.847	0.960	1.186	-0.024	2.235	1.973	0.995	-0.780	2.059	2.047
Interest payments	0.941	0.944	0.581	1.533	0.638	0.766	0.654	0.466	0.662	0.780	0.780	0.498
Debt payments to debt outstanding	0.254	0.184	0.562	0.157	0.153	0.026	0.126	0.332	0.134	0.000	0.093	0.274
Debt payments to income	0.046	0.051	0.038	0.030	0.026	0.015	0.036	0.034	0.021	0.000	0.035	0.034

Table 5: baseline, PPD, and FPD economies.

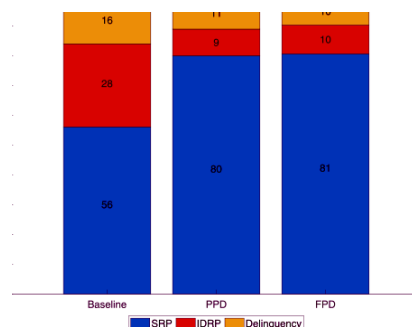


Figure 2: repayment choices.

5.5 Table 6 + Figure 3: consumption, wealth, savings, and debt

Read:

- Table 6 shows that PPD and FPD increase consumption early in life and shift debt repayment later.
- For ages 26–30, average consumption rises from 29.929 in the baseline to 31.067 under PPD and 31.573 under FPD.
- Debt outstanding remains higher under the deferral contracts because repayment is delayed.
- Figure 3 shows consumption gains early in life and modest reductions later in life.

- The saving response is strong early on, especially under FPD.

Interpretation:

- This table is the mechanism table. Welfare rises because consumption is moved from later, lower-marginal-utility periods to earlier, higher-marginal-utility periods.
- Higher debt under PPD and FPD is not automatically bad in the model. It is the counterpart of better consumption smoothing. The welfare calculation asks whether the extra debt burden later is outweighed by early-life liquidity relief.
- Figure 3 helps separate two channels: consumption increases early, and savings also increase early. The savings response means households do not spend all the freed-up liquidity; they also build precautionary buffers.
- The later-life consumption decline is important because it reveals the trade-off. The policies are not gifts; they are intertemporal reallocations that make payments better aligned with income and wealth.
- FPD has larger early consumption and saving effects than PPD because it removes both principal and interest payments during the first ten years.

What not to over-read:

- The table should not be interpreted as saying that higher debt is welfare-improving in general. Higher debt is welfare-improving here only because it buys liquidity when marginal utility is especially high and because the model reprices loans to hold lender NPV fixed.

Economic message: The contracts work by transferring resources toward early adulthood, when marginal utility is high and households are most exposed to income risk. This is the central economic mechanism behind the welfare gains.

Table 6
Consumption, net wealth and debt outstanding over the lifecycle: This table shows average consumption, net wealth, and debt outstanding over the lifecycle. The first three columns show the statistics for the baseline case, the middle three columns for the PPD policy, and the last three columns for the FPD policy.

Age	Baseline			Principal Payment Deferral (PPD)			Full Payment Deferral (FPD)		
	Cons.	Net wealth	Debt outs.	Cons.	Net wealth	Debt outs.	Cons.	Net wealth	Debt outs.
26-30	29.929	-13.755	26.640	31.067	-14.637	28.668	31.573	-15.494	29.514
31-35	44.635	12.143	16.903	45.211	8.537	28.685	45.525	5.779	33.093
36-40	58.393	52.352	6.572	57.986	44.676	24.430	57.962	40.369	31.568
41-45	69.255	95.032	4.722	68.748	87.758	12.553	68.508	83.764	20.697
46-50	76.385	126.039	4.602	76.017	123.231	4.257	75.770	118.173	8.915
51-55	80.301	137.052	0.952	80.009	132.203	3.831	79.814	129.349	5.308
56-60	82.794	117.927	0.000	82.537	112.937	3.610	82.388	110.766	4.870
61-65	84.356	68.926	0.000	84.167	67.463	0.706	84.058	66.750	0.942
>66	84.117	9.127	0.000	84.105	9.097	0.000	84.096	9.077	0.000
Entire life-cycle average	74.325	43.974	4.026	74.298	41.723	7.116	74.285	40.200	8.994

Table 6: consumption, wealth, and debt over the lifecycle.

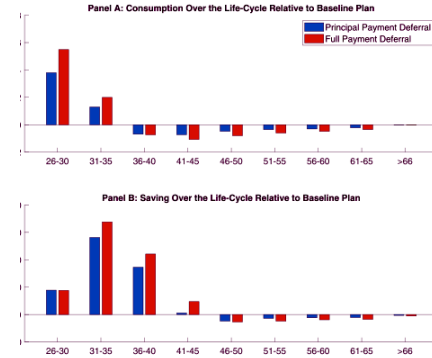


Figure 3: consumption and savings changes.

5.6 Table 7 + Table 8: welfare gains and robustness

Read:

- Table 7 reports annual certainty-equivalent welfare gains of 1.22% for PPD and 2.12% for FPD.
- A simple longer-maturity contract generates only 0.59% welfare gain, much smaller than the deferral contracts.
- Holding the loan premium fixed reduces the PPD gain to 0.48% but leaves the FPD gain relatively high at 1.72%.

- Table 8 shows robustness: gains remain positive under higher lender NPV, different preference parameters, different initial debt levels, unsecured borrowing, and different household structures.
- FPD gains are especially large for borrowers with higher initial debt: 3.35%.

Interpretation:

- Table 7 decomposes why the policies work. The longer-maturity comparison is especially important: simply stretching payments over more years is not the same as removing payments from the first ten years.
- The fixed-premium rows distinguish two channels. For PPD, much of the gain comes from the lower loan premium made possible by lower delinquency. For FPD, the gain remains large even when the premium is held fixed, so the direct liquidity-timing channel is dominant.
- The welfare units are annual certainty-equivalent consumption, so 2% is economically large. It means households would give up about 2% of annual consumption every year to obtain the FPD contract instead of the baseline.
- Table 8 shows that the result is not fragile to one calibration choice. The largest gains appear exactly where the model predicts: households with more initial student debt, more binding borrowing constraints, and stronger early-life liquidity needs.
- Robustness to unsecured borrowing is important because it asks whether private credit could undo the need for deferral. The gains remain positive, implying that outside credit does not fully solve the student-loan timing problem.

What not to over-read:

- Robustness does not mean every omitted real-world feature is irrelevant. It means the main mechanism survives many plausible changes in preferences, borrowing possibilities, and household composition.

Economic message: The welfare gains are large and robust. FPD is most powerful because it fully eliminates early payments, while PPD also benefits from lower loan premia.

Table 7

Welfare gains decomposition: This table reports the welfare gains of the two policies under analysis (PPD and FPD), relative to the baseline case (with the baseline loan premium, $\varphi^{Baseline} = 3.5\%$), as well as the welfare gains of a contract with the same features as the baseline contract but with longer maturity (LM). The different columns report results for different values of the loan premium in the PPD, FPD, and LM economies. In column 2, the loan premium is set at the value obtained in the baseline economy ($\varphi^{Baseline}$), while columns 3, 4 and 5 report the results with the equilibrium loan premium of the PPD, FPD, and LM economies, respectively.

Loan premium (φ)	$\varphi^{Baseline}$	φ^{PPD}	φ^{FPD}	φ^{LM}
Principal Payment Deferral (PPD)	0.48%	1.22%		
Full Payment Deferral (FPD)	1.72%		2.12%	
Longer Maturity (LM)	0.18%			0.59%

Table 7: welfare gains decomposition.

Table 8

Robustness of welfare gains: This table reports the welfare gains for two policies under consideration (PPD and FPD) for different alternative scenarios described in the main text. Panel A reports welfare gains when we require that average loan NPVs are 10% higher than in the baseline (Section 5.5.1). Panel B reports welfare gains for different values of the subjective discount factor and the elasticity of intertemporal substitution (Section 5.5.2). Panel C reports welfare gains when households have ex-ante heterogeneous levels of leverage calibrated using the distribution of starting balances from the Survey of Consumer Finances (Section 5.5.3). Panel D shows the welfare gains when we allow agents to borrow unsecured debt (Section 5.5.4). Panel E reports welfare gains for different household compositions (Section 5.5.5).

Panel A: 10% Higher NPV				
Loan premium (φ)	$\varphi^{Baseline}$	φ^{PPD}	φ^{FPD}	
	3.50%	1.80%		1.56%
PPD	0.48%	1.17%		
FPD	1.72%		2.10%	
Panel B: Different preference parameters				
	Baseline	Lower Beta $\beta = 0.93$	Higher Beta $\beta = 0.97$	Lower EIS $\psi = 0.45$
PPD	1.22%	1.67%	0.85%	1.26%
FPD	2.12%	2.79%	1.52%	2.21%
				Higher EIS $\psi = 0.55$
PPD				1.20%
FPD				2.01%
Panel C: Different initial levels of debt				
	Lower debt	Baseline	Higher debt	
PPD	0.68%	1.15%	1.31%	
FPD	1.15%	2.08%	3.35%	
Panel D: Allowing for unsecured borrowing				
	Baseline $\varphi = +\infty\%$	Unsecured debt $\varphi = 8\%$	Unsecured debt $\varphi = 5\%$	
PPD	1.22%	1.18%	1.14%	
FPD	2.12%	2.03%	1.97%	
Panel E: Different household compositions				
	Baseline	Non-working spouse with children	Single adult with children	Working couple with children
PPD	1.22%	0.78%	0.91%	1.35%
FPD	2.12%	1.75%	1.86%	2.28%

Table 8: robustness of welfare gains.

5.7 Table 9 + Table 10: extensions and policy comparison

Read:

- Table 9 shows that allowing endogenous stock-market participation raises welfare gains to 1.52% under PPD and 2.56% under FPD.
- With prepayment, the welfare gains are 1.08% for PPD and 2.02% for FPD, close to the benchmark values.
- Table 10 compares deferral to Biden-administration policy proposals.
- The debt-forgiveness equivalent is \$10,800 for PPD and \$16,750 for FPD.
- Shortening the IDRFP forgiveness horizon to 10 years produces only a 0.35% welfare gain, while lowering IDRFP payments generates a 1.82% welfare gain.

Interpretation:

- The participation extension in Table 9 matters because deferral can change broader household financial behavior. When early liquidity improves, households may participate more in risky asset markets, raising welfare beyond the baseline consumption-smoothing channel.

- The prepayment rows are a useful discipline. If borrowers can voluntarily repay faster, the welfare gains do not disappear. This means the contract is not valuable only because it forces everyone into delayed repayment; it is valuable because it gives constrained borrowers an option not to repay early.
- Table 10 converts abstract welfare numbers into policy magnitudes. Saying FPD is worth about \$16,750 in debt forgiveness makes the result easier to compare with real proposals.
- The IDRPs comparison is informative. Shorter forgiveness alone gives a small welfare gain because it mainly affects the tail of the repayment horizon. Lower required payments generate larger gains because they directly relax near-term liquidity constraints.
- The policy comparison reinforces the paper’s main message: liquidity timing can mimic some of the welfare benefits of forgiveness without mechanically transferring the same amount of wealth upfront.

What not to over-read:

- The forgiveness equivalence is a model-based welfare equivalence, not a statement that the fiscal costs are the same. The whole point is that deferral can deliver similar utility gains while preserving lender NPV in the model.

Economic message: Payment deferral delivers welfare gains comparable to large debt forgiveness but without the same fiscal transfer. Reforms that lower payments can also be valuable because they operate through the same liquidity channel.

Table 9

Welfare gains with endogenous stock market participation decision and student debt prepayment: Panel A of this table reports moments of the model when agents can endogenously decide to participate in the stock market. The first row of this panel reports the welfare gain. The bottom four rows of Panel A report age, debt outstanding, income and net wealth, on the period agents decide to enter the stock market. Panels B and C of the table show the results of the model when agents can choose to prepay their student debt. The first row of Panel B shows the probability that agents enroll in SRP without prepaying any principal, the probability that they enroll in SRP and prepay some principal, and the probability of enrolling in IDRP or become delinquent. The remaining rows of the panel show income, income growth, leverage, savings, and total payments associated with any of those decisions. Panel C shows the welfare gains of our two policies, PPD and FPD, when prepayment is allowed.

Panel A: Stock market participation decision				
	Baseline	PPD	FPD	
Welfare gain	n/a	1.52%	2.56%	
Decision to participate:				
Age	28.753	28.440	28.309	
Debt outstanding	28.384	29.820	30.636	
Income	59.594	57.889	57.334	
Net wealth	-10.782	-12.232	-13.356	
Panel B: Prepayment drivers				
	SRP	SRP/Prepay	IDRP	Default
Probability	0.186	0.174	0.400	0.240
Income	65.221	87.360	44.316	21.114
Income growth	0.128	0.170	0.060	-0.015
Leverage outstanding	25.984	15.880	25.331	35.271
Savings	16.524	14.926	15.753	6.102
Total payment	3.590	11.829	2.961	0.000
Panel C: Welfare gains PPD/FPD with prepayment				
	Baseline	PPD	FPD	
Loan premia	3.50%	1.74%	1.50%	
Welfare gains		1.08%	2.02%	

Table 9: participation and prepayment extensions.

Table 10

Comparison to Biden administration’s policy proposals: This table reports comparisons between the Biden administration’s proposed student debt repayment policies and the baseline, PPD, and FPD models. Panel A reports the amount of student debt forgiveness required to obtain the same welfare gain as the PPD and FPD plans. Panel B reports welfare gains of the two proposed IDRPs plan modifications relative to the baseline.

Panel A: Loan forgiveness		
	PPD	FPD
Welfare-equivalent debt reduction	\$10,800	\$16,750
Panel B: IDRPs Modifications		
	Welfare gain	Loan premium
(i) Shorter repayment period	0.35%	8.86%
(ii) Smaller payments	1.82%	2.77%

Table 10: Biden-administration proposals.

6 Short summary

- The paper builds a lifecycle model of student loan repayment, consumption, saving, and delinquency.
- The baseline economy contains SRP, IDRP, and delinquency choices.
- PPD delays principal payments for the first ten years; FPD delays all payments for the first ten years.
- The model is recalibrated and solved in each counterfactual equilibrium, including loan premium repricing.
- PPD reduces delinquency by roughly one-third; FPD reduces it by roughly one-half.
- Welfare gains are 1.22% per year for PPD and 2.12% for FPD.
- Longer maturity alone delivers much smaller gains because it does not sufficiently reduce the early-life payment burden.
- The gains come from improved consumption smoothing and better insurance against income shocks.
- Loan repricing due to lower delinquency is important for PPD but less important for FPD.
- Robustness exercises confirm that gains persist across preferences, debt levels, unsecured borrowing, household structures, stock-market participation, and prepayment options.
- The welfare gains are comparable to debt forgiveness of \$10,800 for PPD and \$16,750 for FPD.
- The policy message is that targeted payment deferral can deliver large liquidity benefits without the fiscal cost of blanket forgiveness.

7 Conclusion

7.1 Core conclusion

Boutros, Clara, and Gomes show that the timing of student loan payments is central to borrower welfare. Current contracts force payments early in the lifecycle, when borrowers have low wealth, low income, and high marginal utility. Deferring payments generates large welfare gains even when lenders are kept whole.

7.2 Economic intuition

Student debt is unusual because it is non-dischargeable, held early in life, and tied to a lifecycle income path that rises with age. A young borrower forced to repay immediately sacrifices high-value consumption and precautionary saving. Deferral shifts repayments to a period when the borrower can better afford them, lowering delinquency and improving consumption smoothing.

7.3 Takeaway

The paper reframes student-loan policy away from pure debt cancellation. If the policy goal is to help borrowers at the moment when debt is most burdensome, contract design can matter as much as the debt level itself. PPD and FPD deliver welfare gains similar to forgiveness while preserving budget neutrality in the model.